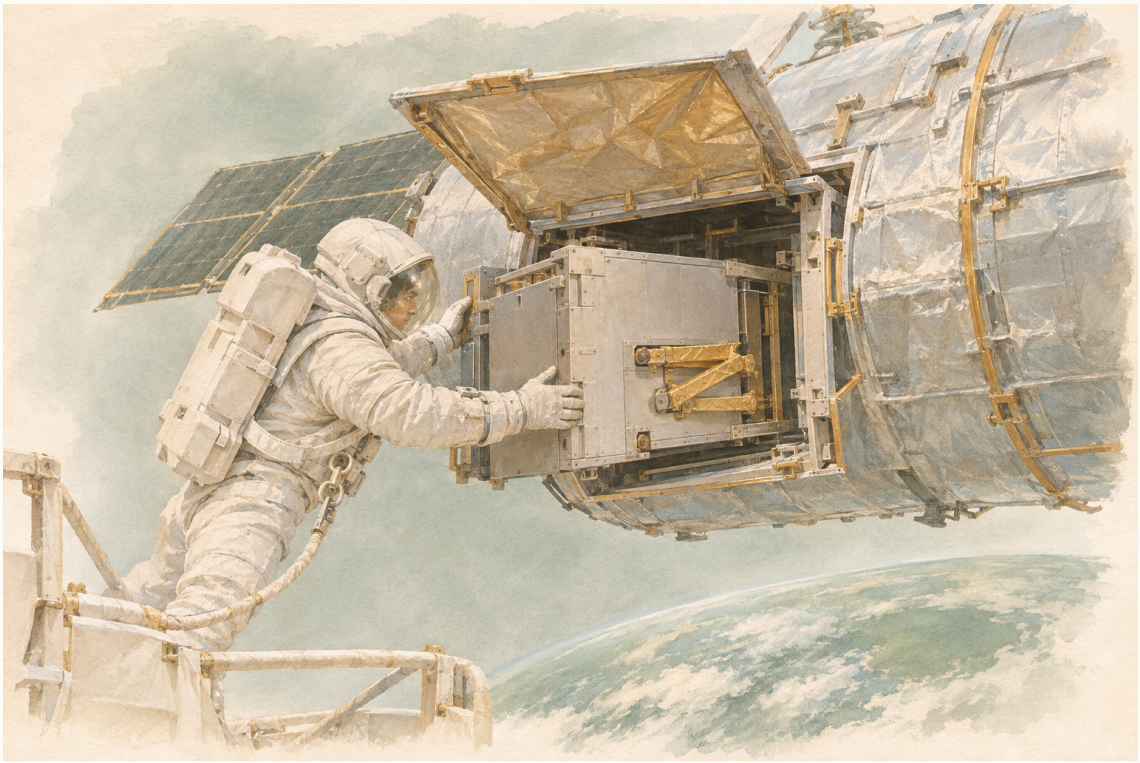
需要戴眼镜的望远镜

The Telescope That Needed Glasses



主镜没有修好，影像怎样变清晰？

READ 纯净阅读 | CHECK 中文核对 | LEARN 词汇记忆 | RECALL 回想巩固

英文原文

On 24 April 1990 the space shuttle Discovery lofted the Hubble Space Telescope into orbit. The observatory cradled in the payload bay had cost about $1.5 billion, a lavish investment meant to fix its gaze on the cosmos and its galaxies. Money, it soon emerged, could not buy clarity.

Its first snapshots came back blurred and fuzzy. In June 1990 NASA announced a flaw in the main mirror: its outer rim had been flattened by 2.2 micrometres, roughly a fiftieth of the width of a hair. That deviation marred the optics, leaving light without a sharp focus.

The trouble began with calibration. A lens in the assembly that checked the mirror's shape had been offset by about 1.3 millimetres, so the mirror was ground to the wrong curve. A second, independent test by the contractor, Perkin-Elmer, gave a divergent result, but engineers and NASA managers judged it less reliable and left the contradictory reading unexplored.

The mirror could not be repaired in orbit, so engineers prescribed tailored glasses. COSTAR, a corrective cabinet the size of a refrigerator, carried small mirrors on adjustable arms that intercepted the light and compensated for the imperfect mirror's distortion before the light reached the instruments. The mirror itself would stay untouched, and a new main camera was prepared.

On 2 December 1993 Endeavour ascended with seven astronauts. After a rendezvous, they worked in tandem across five spacewalks, about 35 and a half hours of exertion, installing COSTAR and the new camera. The imagery turned crisp, Hubble's vision renewed. The mirror still carries that 2.2-micrometre flaw, and the light it gathers is corrected only after leaving it.

中文译文

1990年4月24日，发现号航天飞机将哈勃空间望远镜送入轨道。这座安放在载荷舱中的太空天文台耗资约15亿美元，是一笔用于凝视宇宙与星系的巨额投资。但人们很快发现，花钱并不一定能买来清晰的影像。

哈勃传回的首批照片模糊不清。1990年6月，美国航天局宣布主镜有缺陷：外缘磨得过平，与正确形状相差2.2微米，约为头发丝宽度的五十分之一。这点偏差损害了光学系统，使光线无法准确聚焦。

问题出在校准上。检测主镜形状的装置中，一块透镜错位了约1.3毫米，主镜因此被磨成了错误的曲面。承包商珀金-埃尔默公司另一项独立测试曾给出不同结果，但工程师和美国航天局管理人员认为该测试较不可靠，没有深入追查那组相互矛盾的读数。

主镜无法在轨道上修复，于是工程师为它设计了一副“定制眼镜”。矫正装置COSTAR的箱体约有冰箱大小，可调机械臂上装着小反射镜，在光线进入仪器前将其截住，抵消有缺陷的主镜造成的畸变。主镜本身保持不动，同时还准备了一台新的主相机。

1993年12月2日，奋进号载着七名宇航员升空。完成交会后，他们协同作业，通过五次太空行走，用约35个半小时安装了COSTAR和新相机。哈勃的影像恢复清晰，仿佛重获视力。但主镜那2.2微米的缺陷仍然存在，光线是在离开主镜之后才得到矫正的。

2.2微米是主镜形状误差，1.3毫米是检测装置透镜的错位；安装矫正装置并未修磨主镜。

核心词汇

先记这22项。连着下方的原文短语一起回想。

observatory n.
天文台
The observatory

payload n.
有效载荷
in the payload bay

lavish adj.
耗资巨大的
a lavish investment

clarity n.
清晰度
could not buy clarity

flaw n.
缺陷
a flaw in the main mirror

deviation n.
偏差
That deviation marred the optics

optics n.
光学系统
marred the optics

calibration n.
校准
The trouble began with calibration

assembly n.
装置；组件
A lens in the assembly

offset v.
使偏离位置
offset by about 1.3 millimetres

contractor n.
承包商
the contractor, Perkin-Elmer

divergent adj.
不同的；不一致的
a divergent result

reliable adj.
可靠的
judged it less reliable

contradictory adj.
相互矛盾的
the contradictory reading

corrective adj.
矫正的
a corrective cabinet

intercept v.
截住
intercepted the light

compensate for 动词短语
弥补；抵消
compensated for

distortion n.
畸变
the imperfect mirror's distortion

untouched adj.
未改动的
The mirror itself would stay untouched

rendezvous n.
交会
After a rendezvous

in tandem 介词短语
协同地
worked in tandem

imagery n.
影像
The imagery turned crisp

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

loft v.
向高处送出

telescope n.
望远镜

into prep.
进入；到……中

cradle v.
稳妥安放

investment n.
投资

gaze n.
凝视；目光

cosmos n.
宇宙

galaxy n.
星系

snapshot n.
照片；快照

blurred adj.
模糊的

fuzzy adj.
模糊不清的

outer adj.
外侧的

rim n.
边缘

flatten v.
磨平；使变平

width n.
宽度

mar v.
损害

sharp adj.
清晰锐利的

lens n.
透镜

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

millimetre n.
毫米

be ground to 动词短语
被磨成

curve n.
曲面；曲线

independent test n.
独立测试

unexplored adj.
未被深入调查的

prescribe v.
开出；提出（方案）

tailored adj.
量身定制的

cabinet n.
箱体

refrigerator n.
冰箱

adjustable adj.
可调节的

imperfect adj.
有缺陷的

ascend v.
升空；上升

spacewalk n.
太空行走

exertion n.
费力；辛苦作业

install v.
安装

crisp adj.
清晰鲜明的

vision n.
视力

renew v.
使恢复

gather v.
收集

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The trouble began with _____.

02 engineers and NASA managers judged it less _____ and left the contradictory reading unexplored.

03 The mirror itself would stay _____, and a new main camera was prepared.

04 The _____ turned crisp, Hubble's vision renewed.

文章主线

- 失焦 1990年，耗资约15亿美元的哈勃望远镜升空，照片却模糊；主镜外缘磨错2.2微米，无法准确聚光。
- 根源 主镜被磨错，是因检测装置的透镜错位约1.3毫米；另一独立测试读数不同，却被认为较不可靠而未追查。
- 方案 主镜无法在轨修复，工程师设计矫正装置COSTAR，用小反射镜在光线进入仪器前抵消畸变，并另备新相机。
- 修复 1993年，七名宇航员在五次太空行走中协同作业约35.5小时，装上COSTAR和新相机。
- 结果 影像恢复清晰，但主镜本身始终未动，2.2微米缺陷依旧；被矫正的是离开主镜后的光线。

参考答案　1 calibration 2 reliable 3 untouched 4 imagery